

Follow-up note: empirical triangulation of (r_e/r_0)

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Follow-up note: empirical triangulation of r_e/r_0 — for Tepper Gill

Date: 2026-05-26. **From:** Trey Morris (with Claude Opus 4.7). **Re:** Author guidance recorded after the 2026-05-25 phone call following the interim report¹; empirical triangulation closes the joint-fit thread²; the first-principles rederivation continues separately³.

Tepper — following your guidance that the branches (b) and (c) in §5 of the interim report⁴ are bracketing guides from a uni-observable search rather than theoretical predictions, we have performed the joint fit across all six precision observables that you suggested. We summarise here what we found. The full notebook and per-observable diagnostics are recorded separately⁵.

1. Triangulated value

We obtain, from a precision-weighted χ^2 joint fit,

$$r_e/r_0 = 0.499\,420\,509\,912\,831\,7 \pm 2.5 \times 10^{-13}, \quad (1)$$

where the uncertainty is the one-sigma value taken from the second derivative of χ^2 at the optimum.

We observe that (1) differs from the published $r_e/r_0 = 0.499\,857\,150\,068\,631$ in the fourth decimal place. It is, however, indistinguishable from the back-fit-to- g_s value (branch (c) of the interim report) to sixteen significant figures. The triangulation thus makes quantitative what the Bethe–Salpeter campaign already documented structurally⁶. Indeed, every g_s -dependent observable in the framework satisfies

¹Interim report, pull request #59 (github.com/temoTxt/PyPhysics/pull/59). §5 of that report sketches the branches (b) and (c) discussed here.

²Issue #61 (github.com/temoTxt/PyPhysics/issues/61). Empirical triangulation thread.

³Issue #54 (github.com/temoTxt/PyPhysics/issues/54). First-principles rederivation thread.

⁴Interim report, pull request #59 (github.com/temoTxt/PyPhysics/pull/59). §5 of that report sketches the branches (b) and (c) discussed here.

⁵Roadmapping/Mathematica_Notebooks/Quantum_Mechanics/ r_e _triangulation.wl. Joint-fit notebook with per-observable diagnostics.

⁶Roadmapping/Quantum_Mechanics/Bethe_Salpeter/10_CrossComparison.md, §2 (the r_e back-fit self-consistency across six g_s -dependent observables).

$$\text{prediction} = (g_s/ - 2)^n \times \text{textbook}, \quad n \in \{1, 2\}, \quad (2)$$

so a single back-fit applied to six observables yields a single r_e value.

2. Per-observable residuals at the triangulated optimum

The fit residuals are collected in Table 1.

Observable	Prediction	Measurement	Residual	Source of residual
Electron g_s	$-2.00231930 \dots$	$-2.00231930 \dots$	0	matches by construction
H $2P_{3/2} - 2P_{1/2}$	10,962 MHz	10,969.13 MHz	-7.13 MHz	Bethe-estimate floor ⁷
H 1S hyperfine	1,420.04 MHz	1,420.4058 MHz	-0.366 MHz	sub-leading QED ⁸
He $^3P_0 - ^3P_1$	29,616.95 MHz	29,616.952 MHz	-0.002 MHz	kHz floor ⁹
Positronium	203,389 MHz	203,389 MHz	0	matches by construction
ortho-para Muonium hyperfine	4,463.40 MHz	4,463.3028 MHz	$+0.097$ MHz	sub-leading QED ¹⁰

Table 1. Per-observable residuals at the triangulated optimum (1).

Every non-zero residual in Table 1 is a documented framework-precision-floor effect. It is not reducible by re-tuning r_e/r_0 . No observable is in tension with (1) once the framework’s known precision scope is respected.

3. Honest scoping note on the objective function

We record one substantive-AI decision exposed by the joint fit. The literal “fit to all six measurements” reading treats each measurement’s stated uncertainty as the σ_i entering χ^2 . Under that reading the optimum is

$$r_e/r_0 = 0.499\,406\,1\dots, \quad (3)$$

differing from (1) in the fifth decimal place. Equation (3) pulls g_e off the measured value by approximately

⁷Roadmapping/Quantum_Mechanics/Bethe_Salpeter/14_Lamb_Shift_2P.md, §14.2 (Bethe-estimate precision floor).

⁸Roadmapping/Quantum_Mechanics/Bethe_Salpeter/22_Hydrogen_Hyperfine.md, §22.1 (sub-leading QED gap).

⁹Roadmapping/Quantum_Mechanics/Bethe_Salpeter/07_Helium_Fine_Structure.md, §72 (kHz precision floor).

¹⁰Roadmapping/Quantum_Mechanics/Bethe_Salpeter/08_Muonium_Hyperfine.md, §80 (sub-leading QED gap).

$$\Delta g_e \approx 5.8 \times 10^{-5}, \quad (4)$$

in an attempt to compensate the hyperfine and helium fine-structure residuals. This is an artifact of treating framework-precision-floor residuals as if they were measurement uncertainty.

The honest formulation adds a framework-floor noise term to each σ_i ,

$$\sigma_i^2 = \sigma_{\text{meas},i}^2 + \sigma_{\text{floor},i}^2. \quad (5)$$

The recognition behind (5) is that the leading $(g_s/ - 2)^n$ model cannot fit Bethe-estimate-gap residuals by any choice of r_e alone. Under the weighting prescribed by (5) the optimum is the value (1), matching branch (c) of the interim report to sixteen significant figures. Both passes are documented in the notebook¹¹. We send you the honest answer here, with the alternative (3) recorded for transparency.

4. Where this leaves the open question

A natural way to read the relationship between the published r_e and the triangulated value follows. The published $r_e/r_0 = 0.499\,857\dots$ is an *initial-value* result from the working-notebook numerical search against g_s alone. The joint fit across six observables is a *refinement calculation* on top of that initial value. The triangulated value (1) supersedes the initial value in the same way that any iteratively-refined numerical result supersedes an earlier estimate. It does not invalidate the derivation that produced the initial value. It refines the cutoff identification using six constraints rather than one.

A further refinement is still open: a first-principles derivation of r_e from the dual-Dirac renormalisation prescription, in the sense of DRQM I §III.D. This is tracked separately¹². The candidate starting points are sketched briefly below, and expanded into one-to-three-paragraph explanations each in a companion note¹³.

By “first-principles” we mean a derivation in which r_e emerges from the dual-Dirac equation’s internal structure, rather than being fixed by inverse-solving against the measured g_e . The output would be a closed-form expression for r_e/r_0 in terms of the framework’s fundamental parameters — functions of α , r_0 , and the structural constants of the dual representation. Such an expression would evaluate numerically to approximately $0.499\,420\,509\dots$ if the empirical refinement holds, or to a different value if the derivation exposes a structure the triangulation has not captured.

We have considered three plausible starting points, and would defer to your judgment on which (if any) is natural for this framework.

1. The proper-time self-energy integral in the dual-Dirac formalism, with r_e emerging as the regulator scale at which mass renormalisation closes consistently.

¹¹Roadmapping/Mathematica_Notebooks/Quantum_Mechanics/r_e_triangularion.wl. Joint-fit notebook with per-observable diagnostics.

¹²Issue #54 (github.com/temoTxt/PyPhysics/issues/54). First-principles rederivation thread.

¹³Roadmapping/Author_Reports/2026-05_re_derivation_candidates_for_gill.md. Companion note expanding the three candidate starting points.

2. A variational determination, with r_e fixed by demanding that the renormalised dual-Dirac equation reproduce the physical electron mass at the cutoff. This is analogous to the way the standard-QED running coupling is fixed by a renormalisation condition.
3. A structural constant of the dual representation — something that the b -factor projection structure or the second-order Dirac decomposition singles out — making r_e/r_0 a derivable framework-internal ratio rather than a renormalisation choice.

Per your 2026-05-25 author guidance, a fourth candidate we had initially considered — the retrieval of an unpublished working-notebook derivation step — is empty. The original cutoff value was a numerical search against g_s alone, with no derivation-step working draft to retrieve.

We have not pursued any of the three. The natural starting point depends on the framework's internal logic better than we can guess at from outside. The thread is non-urgent. The triangulated value (1) already serves as the campaign's current best refinement, and a first-principles derivation can sit indefinitely as a would-be-nice-to-have until you point us at a starting structure, or decide the thread is not where the project's time is best spent.

Whichever route turns out to be natural, the outcome could agree with the triangulated value (confirming this refinement), supersede it (a third refinement), or expose a derivation-level structure that reframes the cutoff entirely. Each of these outcomes is consistent with treating each calculation as an iterative refinement rather than as a final authoritative answer.

References